Results: recovery checks and study suite

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Results: recovery checks and study suite

Every quantitative assertion in this and the following results sections is a generated token,

Results: recovery checks... (part 2)

hydrated by the manuscript-variable generator from analysis outputs produced by `src/analysis/workflow.py` and `src/fedference/experiments/` — no number is transcribed by hand.

Results: recovery checks... (part 3)

The studies implement categorical source-mechanism analogues of the colony belief-sharing scenario (Friston et al. 2024) and add the contaminated-sentinel robustness sweep and the federated neural-network baseline that are this paper's robust-federated-learning contribution.

Results: recovery checks... (part 4)

All runs are deterministic under seed 0.

Results: recovery checks... (part 5)

We lead with the recovery limits, not with a study, because they are what makes the studies a single coherent system rather than a collection of unrelated experiments. The generalized-Bayes machinery of sec. 8 contains the standard-Bayes client corner,

Results: recovery checks... (part 6)

while the server has the exact project-local zero-robustness log-linear-pool identity.

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Results: recovery checks... (part 7)

Under the explicit bridge in sec. 8.8, that pool is a categorical specialization of Eq. 7's message-combination term rather than the complete source protocol. We verify those limited identities to machine precision before reporting anything built on top of them.

Recovery limits: standard-Bayes and project-pool corners are exact to machine precision

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Recovery limits (part 2)

The identities that anchor every result are the client recovery of standard Bayes at the KL/NLL/$\beta \to 0$ and $q_{\text{loss}} \to 0$ limits plus the project-local server recovery to the log-linear pool at $c = 0$ — the scoped claims of sec. 9 (Corollary 6,

Recovery limits (part 3)

Lemma 3, Theorem 5).

These are not figures but exact equalities, pinned by the locked core test suite.

Recovery limits (part 4)

Under the theorem's shared-support, posterior-log-potential, and fixed-weight assumptions, the server pool specializes Eq. 7's message-combination term; it does not reproduce the source construction in full. Robustness is a tested extension that vanishes at the stated recovery limits.

Recovery limits (part 5)

The five residuals below are the maximum absolute deviations between each generalized-Bayes object and the standard object it must reproduce in the trusting limit.

Recovery limits (part 6)

Each is a deterministic constant of the mathematics, not a per-run sample: it is reported as the maximum absolute deviation over the recovery band, which is exactly 0 where the implementation evaluates the closed form at the limit (the Rényi/loss switch) and otherwise a tiny floating-point residual:

Recovery limits (part 7)

Server recovery. The server-side aggregator at zero robustness equals the log-linear pool (eq. 10, Theorem 5): maximum absolute deviation 0. This is the *naive-recovery* limit of the server-side heuristic —

Recovery limits (part 8)

the only property proven for that axis (see sec. 6.3).

Recovery limits (part 9)

Posterior recovery. The generalized posterior under the KL divergence and the NLL loss equals the closed-form prior$\times$likelihood Bayes posterior (eq. 11, Corollary 6): maximum absolute deviation 5.55e-17.

Recovery limits (part 10)

Divergence recovery. The Rényi divergence recovers KL as $\alpha \to 1$ (eq. 6, Lemma 3): residual 0.

Recovery limits (part 11)

Loss recoveries. The β-loss recovers the NLL as $\beta \to 0$ (eq. 7, Proposition 4): residual 0. The robust categorical cross-entropy recovers the NLL as $q_{\text{loss}} \to 0$ (eq. 8,

Recovery limits (part 12)

Proposition 4): residual 0.

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Recovery limits (part 13)

Because the Rényi divergence and the two categorical losses switch to their exact closed form inside narrow numerical-stability bands around the limit point (the Rényi switch band for α and the categorical-loss switch band for q_{loss} and β),

Recovery limits (part 14)

the three zero residuals above confirm that branch equals the standard object — not, by themselves, that the *general* formula converges there.

Recovery limits (part 15)

As a genuine (non-branch) convergence witness, evaluating each general formula strictly *outside* its switch band — $q_{\text{loss}} = \beta = 1.00 \times 10^{-6}$ for the categorical losses and $\alpha = 1.00001$ for the Rényi divergence — gives residuals 1.12e-05 (rcce), 1.24e-05 (β-loss), and 1.66e-05 (Rényi).

Recovery limits (part 16)

These residuals are nonzero—a small multiple of the input offset itself, as the first-order Taylor behavior near the limit predicts—yet remain several orders of magnitude below the $O(1)$ scale of the loss/divergence values being compared, and shrinking monotonically as the offset shrinks toward the switch band.

Recovery limits (part 17)

The verification is in `test_core_identities.py` under `tests/fedference/`.

This is evidence that the general formula itself converges to the standard-Bayes limit, not merely that the implementation switches to it exactly at the corner.

Recovery limits (part 18)

The first residual is the naive-aggregate limit of the *server-side* heuristic (Theorem 5);

Recovery limits (part 19)

the latter four are the per-agent generalized-Bayes recoveries (Corollary 6 + Proposition 4) and the divergence-family recovery in the Rényi limit (Lemma 3,

Recovery limits (part 20)

eq. 6) that define the theorem-bearing FedGVI axis under matching assumptions.

Recovery limits (part 21)

Keeping the three axes distinct at the level of the recovery limits is what lets the robustness claims of sec. 10.5 and sec. 10.6 rest on the per-agent axis without leaning on the heuristic.

Recovery limits (part 22)

283 of 287 acceptance criteria are verified. The pure-NumPy/SciPy core carries project test coverage of 92.47% (gate $\geq 90\%$), with every stochastic step threaded through a single seeded `np.random.default_rng(0)`. sec. 13 records the full environment fingerprint,

Recovery limits (part 23)

and the expected-free-energy identity that underwrites the active-inference substrate is proven and visualized in sec. 9.2 (fig. 11).